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Micro-light-emitting diode display panel

Abstract

A micro-light-emitting diode (microLED) display panel includes microLEDs disposed on a display area, which is divided into a plurality of blocks; drivers respectively disposed on the blocks; and probe pads disposed within each block, the probe pads being electrically connected to corresponding microLEDs via corresponding electric wires. The probe pads and the electric wires of a corresponding block are disposed locally and confined to the corresponding block, and are not connected elsewhere beyond the corresponding block.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/270,164, filed on Oct. 21, 2021, the entire content of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention generally relates to a micro-light-emitting diode (microLED) display panel, and more particularly to a microLED display panel with local probe pads.

2. Description of Related Art

(2) A micro-light-emitting diode (microLED, mLED or μ LED) display panel is one of flat display panels, and is composed of microscopic microLEDs each having a size of 1-100 micrometers. Compared to conventional liquid crystal display panels, the microLED display panels offer better contrast, response time and energy efficiency. Although both organic light-emitting diodes (OLEDs) and microLEDs possess good energy efficiency, the microLEDs, based on group III/V (e.g., GaN) LED technology, offer higher brightness, higher luminous efficacy and longer lifespan than the OLEDs.

(3) As microLEDs are individually transferred to and bonded on a glass substrate to form a display panel, defects may sometimes occur and repairs should be carried out to correct them. Therefore, detection becomes indispensable to manufacturing the microLED display panels.

(4) Probe pads are conventionally set on peripheral areas on the display panels for performing

detection on the bonded microLEDs. However, the microLEDs may probably be overdriven or under driven due to impedance being increased along electric wires. Further, it is ordinarily difficult to identify locations of abnormality due to large amount of microLEDs under detection being electrically connected.

(5) A need has thus arisen to propose a novel scheme to overcome drawbacks of detecting conventional microLED display panels.

SUMMARY OF THE INVENTION

(6) In view of the foregoing, it is an object of the embodiment of the present invention to provide a microLED display panel with locally disposed probe pads facilitating independent detections on local areas respectively, thereby effectively enhancing performance of detection and repairing.

(7) According to one embodiment, a micro-light-emitting diode (microLED) display panel includes microLEDs, drivers and probe pads. The microLEDs are disposed on a display area, which is divided into a plurality of blocks. The drivers are respectively disposed on the blocks. The probe pads are disposed within each block, the probe pads being electrically connected to corresponding microLEDs via corresponding electric wires. The probe pads and the electric wires of a corresponding block are disposed locally and confined to the corresponding block, and are not connected elsewhere beyond the corresponding block.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a schematic diagram illustrating a top view of a micro-light-emitting diode (microLED) display panel according to one embodiment of the present invention;

(2) FIG. 2 shows a schematic diagram illustrating a top view of the microLED display panel with an enlarged block with details according to one embodiment of the present invention;

(3) FIG. 3A shows a simplified schematic diagram illustrating a top view of a block according to a first exemplary embodiment of the present invention;

(4) FIG. 3B shows a simplified schematic diagram illustrating a top view of a block according to a second exemplary embodiment of the present invention;

(5) FIG. 3C shows a simplified schematic diagram illustrating a top view of a block according to a third exemplary embodiment of the present invention;

(6) FIG. 4A shows a block diagram illustrating an integrated machine composed of a microLED chip bonding system, a microLED detecting system and a microLED repair system according to one embodiment of the present invention;

(7) FIG. 4B shows a block diagram illustrating an integrated machine composed of a microLED chip bonding system and a microLED detecting system;

(8) FIG. 4C shows a block diagram illustrating an integrated machine composed of a microLED detecting system and a microLED repair system;

(9) FIG. 4D shows a block diagram illustrating an integrated machine composed of a microLED chip bonding system and a microLED repair system; and

(10) FIG. 5A and FIG. 5B show schematic diagrams respectively illustrating top views of microLED display panels without adopting aspects of the embodiments of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(11) FIG. 1 shows a schematic diagram illustrating a top view of a micro-light-emitting diode (microLED) display panel **100** according to one embodiment of the present invention. Specifically, the microLED display panel **100** may include a display area **101** that is arranged within a perimeter of the microLED display panel **100**. Accordingly, an area between the perimeter of the microLED display **100** and the display area **101** represents a non-display area (or periphery area).

(12) In the embodiment, the display area **101** may be divided into a plurality of blocks **11**, and each

block **11** may have a corresponding driver **12** for driving corresponding microLEDs **13**, for example, red microLEDs **13R**, green microLEDs **13G** and blue microLEDs **13B**, as demonstrated in the magnified view of a block **11** in FIG. **1**. It is noted that, in the specification, the division among the blocks **11** indicates arranging areas for corresponding blocks **11**, instead of physically cutting the microLED display panel **100** and physically separating the blocks **11** from each other. The microLED display panel **100** of the embodiment may be, but not limited to, a full-color display panel. It is appreciated that the microLEDs need not be red microLEDs **13R**, green microLEDs **13G** and blue microLEDs **13B** inherently. Instead, for example, the microLEDs may be inherently monochrome but can emit color light by adding fluorescent powder or quantum dots.

(13) FIG. **2** shows a schematic diagram illustrating a top view of the microLED display panel **100** with an enlarged block **11** with details according to one embodiment of the present invention. According to one aspect of the embodiment, each block **11** may include a plurality of (local) probe pads **14**, for example, a reference probe pad **14C**, a red probe pad **14R**, a green probe pad **14G** and a blue probe pad **14B**, which are disposed on the corresponding block **11**. Each block **11** may include a plurality of (local) electric wires **15**, for example, a reference electric wire **15C**, a red electric wire **15R**, a green electric wire **15G** and a blue electric wire **15B**, which are disposed on the corresponding block **11**. In each block **11**, the probe pads **14** are electrically connected to corresponding microLEDs (not shown) via corresponding electric wires **15**. Accordingly, after bonding microLEDs on the block **11** of the display area **101**, detection may be performed on the microLEDs of the block **11** by providing or receiving signals (or voltages) to or from the probe pads **14**. For example, a reference voltage V_{com} may be provided to or received from the reference probe pads **14C**, and a color (e.g., red, green or blue) signal may be provided to or received from the (red, green or blue) probe pad (**14R/14G/14B**). Repairs may be executed afterward if abnormality is detected.

(14) It is appreciated that the probe pads **14** (of each block **11**) may be disposed (or located) in a variety of places. For example, the probe pads **14** may be disposed on an edge of (but within) the block **11**, or be disposed on one or two sides of the driver **12**. The probe pads **14** (of each block **11**) may be arranged in a variety of ways. For example, the probe pads **14** may be arranged laterally, longitudinally or individually (or separately).

(15) According to another aspect of the embodiment, the probe pads **14** and electric wires **15** of a corresponding block **11** are disposed locally, that is, confined to the corresponding block **11**, and are not connected elsewhere beyond the corresponding block **11**. Therefore, detection performed on one block **11** could not affect or be affected by other blocks **11**. Furthermore, locations of abnormality may be definitely identified with certainty. After finishing the detection (and repair, if required) of one block **11**, connection between the probe pads **14** and the microLEDs **13** of that block **11** may be permanently severed, for example, by physically cutting (for example, by laser).

(16) FIG. **3A** shows a simplified schematic diagram illustrating a top view of a block **11** according to a first exemplary embodiment of the present invention. In the embodiment, the probe pads **14** are disposed on an edge of the block **11**. Specifically, same-color bonding pads **16** (for bonding with the microLEDs **13**) at same columns are electrically connected (via longitudinal electric wires **15**), respectively. Next, each column is electrically connected to a corresponding probe pad **14** (via a lateral electric wire **15**). After finishing the detection (and repair, if required) of one block **11**, connection between the probe pads **14** and the microLEDs **13** of the block **11** may be permanently severed, for example, by physically cutting (for example, by laser) as denoted by the dashed arrows. The amount of cuttings is equal to the resolution of the microLED display panel **100**.

(17) FIG. **3B** shows a simplified schematic diagram illustrating a top view of a block **11** according to a second exemplary embodiment of the present invention. In the embodiment, the probe pads **14** are disposed on an edge of the block **11**. Specifically, same-color bonding pads **16** at same columns are electrically connected (via longitudinal electric wires **15**), respectively. Next, each column is electrically connected to a corresponding (longitudinal) compiling wire **17** (via a lateral electric

wire **15**). In the embodiment, there are totally four compiling wires **17**, that is, a reference compiling wire, a red compiling wire, a green compiling wire and a blue compiling wire. The compiling wires **17** are then electrically connected to corresponding probe pads **14** respectively. After finishing the detection (and repair, if required) of one block **11**, connection between the probe pads **14** and the microLEDs **13** of the block **11** may be permanently severed, for example, by physically cutting (for example, by laser) as denoted by the dashed arrows. The amount of cuttings is equal to the number of the blocks **11** in the microLED display panel **100**.

(18) FIG. **3C** shows a simplified schematic diagram illustrating a top view of a block **11** according to a third exemplary embodiment of the present invention. In the embodiment, the probe pads **14** are disposed on sides of the driver **12**. Specifically, same-color bonding pads **16** (for bonding with the microLEDs **13**) at same columns are electrically connected (via longitudinal electric wires **15**), respectively. Next, each column is electrically connected to a corresponding (lateral) compiling wire **17**. In the embodiment, there are totally three compiling wires **17**, that is, a red compiling wire, a green compiling wire and a blue compiling wire. The compiling wires **17** are then electrically connected to corresponding probe pads **14** respectively. After finishing the detection (and repair, if required) of one block **11**, connection between the probe pads **14** and the microLEDs **13** of the block **11** may be permanently severed, for example, by physically cutting (for example, by laser) as denoted by the dashed arrows. The amount of cuttings is at least two times the number of the blocks **11** in the microLED display panel **100**.

(19) FIG. **4A** shows a block diagram illustrating an integrated machine **200A** composed of a microLED chip bonding system **21**, a microLED detecting system **22** and a microLED repair system **23** according to one embodiment of the present invention. After microLEDs are bonded on a block **11**, detection may be immediately performed on that block **11**, followed by repairing defective microLEDs if any defective microLEDs have been detected. Accordingly, process of manufacturing the microLED display panel **100** may be greatly simplified and performance thereof may be substantially increased.

(20) Although the microLED chip bonding system **21**, the microLED detecting system **22** and the microLED repair system **23** are incorporated in the integrated machine **200A** in the embodiment, it is appreciated that, in other embodiments, only two of the three systems may be selected and incorporated in an integrated machine. FIG. **4B** shows a block diagram illustrating an integrated machine **200B** composed of a microLED chip bonding system **21** and a microLED detecting system **22**. FIG. **4C** shows a block diagram illustrating an integrated machine **200C** composed of a microLED detecting system **22** and a microLED repair system **23**. FIG. **4D** shows a block diagram illustrating an integrated machine **200D** composed of a microLED chip bonding system **21** and a microLED repair system **23**.

(21) FIG. **5A** and FIG. **5B** show schematic diagrams respectively illustrating top views of microLED display panels **300A** and **300B** without adopting aspects of the embodiments of the present invention. Specifically, electric wires **31** are routed globally across the entire microLED display panel **300A/300B**, and probe pads **32** are disposed outside the display area **301**. As the impedance of the electric wires **31** is increasing along the length of the electric wires **31**, microLEDs away from the probe pads **32** will be under driven with dimmed lightness. Moreover, as a large amount of microLEDs are driven at the same time, it is difficult to identify locations of abnormality. After finishing detection (and repair, if required), in addition to permanently severing the connection between the probe pads **32** and the display area **301**, all junctions between any neighboring blocks **11** need also be severed permanently, for example, by laser cutting. While performing these laser cuttings, it is inevitable that some fault cuttings may occur and may probably cause the microLED display to malfunction.

(22) Although specific embodiments have been illustrated and described, it will be appreciated by those skilled in the art that various modifications may be made without departing from the scope of the present invention, which is intended to be limited solely by the appended claims.

Claims

1. A micro-light-emitting diode (microLED) display panel, comprising: a plurality of microLEDs disposed on a display area, which is divided into a plurality of blocks; a plurality of drivers respectively disposed on the plurality of blocks; and a plurality of probe pads disposed within each block, the probe pads being electrically connected to corresponding microLEDs via corresponding electric wires; wherein the probe pads and the electric wires of a corresponding block are disposed locally and confined to the corresponding block, and are not connected elsewhere beyond the corresponding block; wherein same-color bonding pads at same columns are connected in the corresponding block, bonding pads of each column are connected to a corresponding compiling wire in the corresponding block before the compiling wire is connected to corresponding probe pad in the corresponding block.
 2. The panel of claim 1, wherein the plurality of microLEDs comprise red microLEDs, green microLEDs and blue microLEDs.
 3. The panel of claim 1, wherein the plurality of microLEDs comprise monochrome microLEDs with added fluorescent powder or quantum dots.
 4. The panel of claim 1, wherein connection between the probe pads and the microLEDs of the corresponding block is permanently severed after finishing detection of the corresponding block.
 5. The panel of claim 4, wherein the connection is severed by laser.
 6. The panel of claim 1, wherein the probe pads are disposed on an edge of the corresponding block.
 7. The panel of claim 1, wherein the probe pads are disposed on sides of the driver.
 8. An integrated machine, comprising and integrating at least two of a micro-light-emitting diode (microLED) chip bonding system, a microLED detecting system and a microLED repair system adapted to a microLED display panel that comprises: a plurality of microLEDs disposed on a display area, which is divided into a plurality of blocks; a plurality of drivers respectively disposed on the plurality of blocks; and a plurality of probe pads disposed within each block, the probe pads being electrically connected to corresponding microLEDs via corresponding electric wires; wherein the probe pads and the electric wires of a corresponding block are disposed locally and confined to the corresponding block, and are not connected elsewhere beyond the corresponding block; wherein same-color bonding pads at same columns are connected in the corresponding block, bonding pads of each column are connected to a corresponding compiling wire in the corresponding block before the compiling wire is connected to corresponding probe pad in the corresponding block.
 9. The machine of claim 8, wherein the plurality of microLEDs comprise red microLEDs, green microLEDs and blue microLEDs.
 10. The machine of claim 8, wherein the plurality of microLEDs comprise monochrome microLEDs with added fluorescent powder or quantum dots.
 11. The machine of claim 8, wherein connection between the probe pads and the microLEDs of the corresponding block is permanently severed after finishing detection of the corresponding block by the microLED detecting system.
 12. The machine of claim 11, wherein the connection is severed by laser.
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